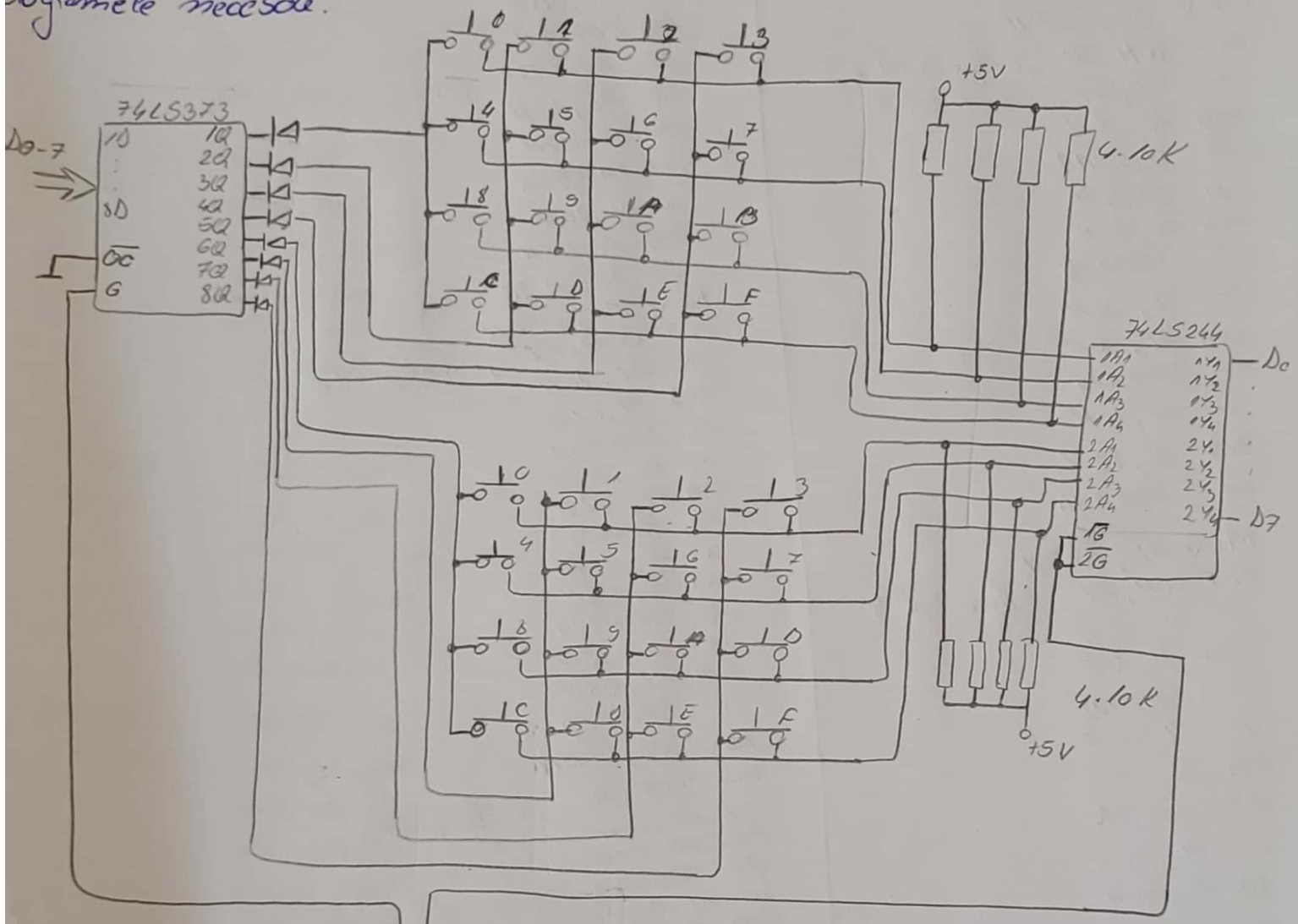
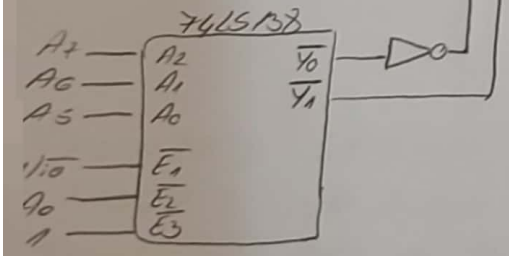
11. Să se conecteze 2 minitastaturi cu câte 16 contacte o UC cu microprocesorul 8086. Se va prezenta schema și programele necesare.





Pentru conectarea tastaturii este necesar un registru și un circuit cu porți cu 3 stări.

Programul:

```
; scanarea tastaturii 1
REIA:   MOV AL, 0DFH
        OUT C0H, AL ; plasez col. test. 1 e pe 0 logic
        IN  AL, 20H ; citim tastatura 1
        AND AL, 10H
        JZ  TASTA10
        IN  AL, 20H
        AND AL, 20H
        JZ  TASTA11
```

```
IN AL, 20H
AND AL, 40H
JZ TASTA18
IN AL, 20H
AND AL, 80H
JZ TASTA1C

MOV AL, 0BFH
OUT 00H, AL ; o logic pe col. 2
IN AL, 20H ; se cit test. 1
AND AL, 16H
JZ TASTA11
IN AL, 20H
AND AL, 20H
JZ TASTA14
IN AL, 20H
AND AL, 40H
JZ TASTA18
IN AL, 20H
AND AL, 80H
JZ TASTA1D

MOV AL, 07FH
OUT 00H, AL ; o logic col. 3
IN AL, 20H
AND AL, 10H
JZ TASTA12
IN AL, 20H
AND AL, 20H
JZ TASTA1C
IN AL, 20H
AND AL, 40H
JZ TASTA1A
IN AL, 20H
AND AL, 80H
JZ TASTA1E

MOV AL, 0DFH
OUT 00H, AL ; o logic pe col 4
IN AL, 20H
AND AL, 10H
JZ TASTA13
IN AL, 20H
AND AL, 20H
JZ TASTA17
IN AL, 20H
AND AL, 40H
JZ TASTA1B
IN AL, 20H
AND AL, 80H
```

; SCANARE TASTATURA 2

```
MOV AL, 0FEH
OUT 00H, AL ; o logic pe primo col
IN AL, 20H ; cit. test. 2
AND AL, 01H
JZ TASTA20
IN AL, 20H
AND AL, 02H
JZ TASTA24
IN AL, 20H
AND AL, 04H
JZ TASTA28
IN AL, 20H
AND AL, 08H
JZ TASTA2C
MOV AL, 0FDH
OUT 00H, AL ; o logic pe col 2
IN AL, 20H
AND AL, 01H
JZ TASTA21
IN AL, 20H
AND AL, 02H
JZ TASTA25
IN AL, 20H
AND AL, 04H
JZ TASTA29
IN AL, 20H
AND AL, 08H
JZ TASTA2D
MOV AL, 0FBH
OUT 00H, AL ; o logic pe col 3
IN AL, 20H
AND AL, 01H
JZ TASTA22
IN AL, 20H
AND AL, 02H
JZ TASTA26
IN AL, 20H
AND AL, 04H
JZ TASTA2A
IN AL, 20H
AND AL, 08H
JZ TASTA2E

MOV AL, 0FFH
OUT 00H, AL ; o logic col 4
IN AL, 20H
AND AL, 01H
JZ TASTA23
IN AL, 20H
AND AL, 02H
JZ TASTA27
IN AL, 20H
AND AL, 04H
JZ TASTA2B
IN AL, 20H
AND AL, 08H
JZ TASTA2F
```

```
TASTA10:  CALL TIM      ; se cst. gd. vibrațiilor
TAS10:    IN AL, 20H
          AND AL, 10H
          JZ TAS10      ; se cst dezactivat tastei
          CALL TIM      ; se cst gd. vibr
          JP REIA

TASTA14:  CALL TIM
TAS14:    IN AL, 20H
          AND AL, 20H
          JZ TAS14
          CALL TIM
          JP REIA

  . . . . .

TASTA1C:  CALL TIM
TAS1C:    IN A, 20H
          AND AL, 80H
          JZ TAS1C
          CALL TIM
          JP REIA

TASTA20:  CALL TIM
TAS20:    IN A, 20H
          AND AL, 01H
          JZ TAS20
          CALL TIM
          JP REIA

  . . . .

TASTA2F:  CALL TIM
TAS2F:    IN A, 20H
          AND AL, 08H
          JZ TAS2F
          CALL TIM
          JP REIA
```